

Optimizing the use of portable ROVs for community-based benthic surveys: A case study on the demographics of sea cucumbers around nursery habitats in the Arctic

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ABSTRACT

Nearshore areas around Qikiqtaaluk in the Canadian Arctic have been identified as nursery habitats for the sea cucumber *Cucumaria frondosa*, which is locally consumed and being explored for small-scale commercial fisheries. This study characterized this species' demographics along a depth gradient (~0.5–11.5 m) at three sites inside and adjacent to nursery grounds using a mini-class remotely operated vehicle (ROV). A novel technique involving the superimposition of virtual quadrats was developed to minimize perspective biases for the analysis of images captured obliquely relative to the seafloor. Overall densities at Katak and Kataaluk were a magnitude lower than at Sanikiluaq, suggesting spatial variability in environmental conditions. Smaller individuals (~5 cm) occurred mostly on rubble between 1 and 3 m at Sanikiluaq, larger ones (~10 cm) on gravel at 2–7 m at Kataaluk, and a mix of both size classes occupied gravel, rubble, boulders, and bedrock between 3 and 11 m at Katak. Globally, body sizes tended to increase with depth, evoking cohorts of sea cucumbers undergoing step-wise downward migrations as they grew. This study provides foundational data on *C. frondosa* around its nursery habitats. It also enhances the usefulness of mini-class ROVs to investigate benthic assemblages in ice-covered areas through community-led initiatives.

1. Introduction

The northern sea cucumber, *Cucumaria frondosa* (Gunnerus, 1767), is a holothuroid echinoderm that is ecologically and commercially important (Gianasi et al., 2021; Mercier et al., 2023). Spawning in *C. frondosa* occurs annually and the fertilized oocytes develop into pelagic lecithotrophic larvae (Hamel and Mercier, 1995, 1996a, 1996b; Mercier et al., 2023). Upon settling, the late pentactula larvae seek shelter in crevices or in complex substrata; the recruits remain cryptic until they are older and ready to move to more exposed areas (Hamel and Mercier, 1996a; Mercier et al., 2023). Throughout early ontogenetic development, as feeding tentacles develop, the posture of *C. frondosa* changes from supine to upright with tentacles extended up in the water column (Gianasi et al., 2018). Diet is concurrently suspected to change from deposited organic matter to plankton that is suspended in the water column (Hamel and Mercier, 1998; Gianasi et al., 2018). Benthic suspension feeders like *C. frondosa* enrich the ecosystem while they feed, as they remove large quantities of suspended particles from the water

column, facilitating nutrient cycling and energy flow (Dame, 1993; Cranford, 2018).

Commercial demand for sea cucumbers originating from the North Atlantic has been steadily growing as a result of the overexploitation of previously targeted species elsewhere in the world (Anderson et al., 2011; Purcell et al., 2013; Mercier et al., 2024). In this context, a fishery for the sea cucumber *C. frondosa* has developed in eastern North America since the 1980s (Gianasi et al., 2021; Mercier et al., 2023). In addition, this species is harvested for subsistence and under consideration for a small-scale fishery in the Canadian Arctic (Hamel and Mercier, 2024). Fisheries for various species of sea cucumber in other parts of the world have been historically unsustainable due to the combination of high value, strong fishing pressure, ineffective management, and biological characteristics that make sea cucumbers vulnerable to overfishing (Purcell et al., 2013; Mercier et al., 2024, 2025). More information is needed about the ecology of *C. frondosa* in its Arctic range to guide sustainable fishing practices and help prevent the boom-and-bust cycle that is common for sea cucumber fisheries.

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In addition to horizontal distribution (geographic range), the vertical distribution (bathymetric range) of a species is an important ecological concept that has been extensively studied (Pineda, 1993; Pineda and Caswell, 1998; Leis and McCormick, 2002; Vedenin et al., 2021). Vertical range could influence the physical and biological conditions a species encounters, such as salinity, food availability, water currents, and predator abundance, which vary with depth (Leis and McCormick, 2002). Ontogenetic migration refers to the movement of individuals of a species through different areas of their habitat as they progress through life stages, from post-settlement to adult (Lincoln et al., 1998). This process can help explain patterns of vertical distribution, particularly in species that transition from a pelagic existence to a benthic one, where habitat requirements may differ substantially. Ontogenetic migrations are largely the result of changes in the ability to sense and respond to environmental cues throughout development and growth (Kingsford et al., 2002; Leis, 2010). The sea cucumber *C. frondosa* has a bipartite (pelago-benthic) life history and newly-settled recruits have only ever been documented in shallow nursery sites where no large individuals were present (Hamel et al., 2023). Dive surveys conducted in the St. Lawrence Estuary in the early 1990s first suggested the possibility of a seasonal or ontogenetic migration based on the number of individuals <30 mm found between 0 and 20 m depths and their gradual disappearance in deeper areas (Hamel and Mercier, 1996a). However, knowledge of nursery grounds, home to newly-settled recruits, did not exist at the time.

To date, the only nursery habitats to have been characterized for *C. frondosa* are located in the shallow nearshore waters around Qikiqtait (traditional name of Belcher Islands, Nunavut) in the Canadian Arctic (Hamel et al., 2023). Juveniles as small as 0.9 mm in length (size of the settling pentactula) were observed at depths from 1.8 to 2.5 m in Kataapik, whereas larger individuals were recorded at greater depths (Hamel et al., 2023). Since large adults were commonly found at high abundance between depths of 30 and 60 m (Hamel and Mercier, 1996a; Gianasi et al., 2021), this initial characterization of a shallow nursery reinforced the hypothesis that individuals migrate to greater depths as they advance through their life stages.

The present study aimed to further explore this assumption by broadly characterizing variations in the size structure and density of populations of *C. frondosa* at different depths, both from within nursery habitats (to ~2.5 m depth) and downslope from them (to ~11.5 m depth). It was assumed that if individuals undergo vertical ontogenetic migration, the body size of individuals should increase with depth as they leave the nursery habitat. Additionally, previous investigations conducted in the Northwest Atlantic suggested that sea cucumbers might migrate to deeper waters upon reaching sexual maturity, possibly due to requirements for more stable environments with age (Hamel and Mercier, 1996a). This is likely the case in Arctic waters (e.g., greater fluctuations in salinity in shallow zones of the water column; Hamel et al., 2023), such that populations of *C. frondosa* were expected to become perceptibly denser at specific depths where environmental conditions are more stable. Challenges associated with surveying shallow benthic environments that are seasonally ice covered underscored the necessity to develop a suitable, cost-effective method that would benefit local community partners. This study thus sought to optimize the use of a maneuverable mini-class remotely operated vehicle (ROV). To quantitatively determine population density using oblique imagery relative to the seafloor, where positional information of the ROV could not be obtained, a technique was developed to superimpose a standardized virtual quadrat on still frames. This technique was validated to rationalize its application within similar community-led studies of remote areas, where dive surveys are impractical or impossible, to support local stewardship of marine resources.

2. Materials and methods

2.1. Study area

To characterize the population of *Cucumaria frondosa*, video-based surveys of the benthic environment were carried in the Hudson Bay, in waters around Qikiqtait, Nunavut, Canada. In May 2023, three sites were surveyed: Sanikiluaq, Katak, and Kataaluk (Fig. 1). Geographically, these sites were between 5.2 and 9.4 km apart by sea. To the southwest of the study area, Sanikiluaq is a shallow shoal at the eastern edge of the hamlet of the same name (56.5482°N, 79.2115°W; Fig. 1). At the time of study, this site was characterized by a near-bottom salinity of 28.5, temperature of -1.1 °C, dissolved oxygen of 101.0 %, and pH of ~7.5 (YSI Pro1030). To the southeast, Katak is a channel about 1 km across (56.5372°N, 79.132764°W; Fig. 1), which had a near-bottom salinity of 29.3, temperature of -1.3 °C, dissolved oxygen of 93.6 %, and pH of 7.6. To the northeast, Kataaluk is another channel about 0.5 km across (56.6033°N, 79.1502°W; Fig. 1) and had a near-bottom salinity of 28.1, temperature of -1.3 °C, dissolved oxygen of 92.8 %, and pH of 7.6. During the study, the site at Sanikiluaq was entirely covered by sea ice and sites at Katak and Kataaluk were mostly covered by ice and had a conspicuous ice edge.

2.2. Video-based benthic surveys

Video-based benthic surveys were made using a portable (mini-class) ROV (Qysea Fifish Pro V6 Plus, Shenzhen, China) that was equipped with a 4K resolution ultra-high-definition camera and two forward-facing LED lights providing 6000 lumens. This ROV unit was launched through the sea ice via an opening or from the ice edge. Visible in the centre of the video footage were two front-facing scale lasers that are 12 cm apart. An average speed of ~0.1 m s⁻¹ was sought. Videos of the benthos were taken with limited control and no record of the camera angle. The overall change in tidal levels during the surveys ranged from 0.03 to 0.19 m and daily tides from the nearest station on the archipelago ranged from 0.43 to 0.67 m (a maximum difference of 0.24 m) above mean lower low water (MLLW). Because tidal variations were ~0.2 m or less, seafloor depth measurements were not adjusted. The maximum seafloor depth varied among the three sites: 3.1 m at Sanikiluaq (shallow shoal), 11.2 m at Katak (channel about 1 km wide), and 7.2 m at Kataaluk (channel about 0.5 km wide).

The substratum was visually characterized using a modified Wentworth classification scale (Platts et al., 1983) with the addition of 'unconsolidated substrata' and 'exposed bedrock' categories, for a total of six types: unconsolidated substrata (size of particle: ≤0.5 cm; e.g., sand, silt, mud, crushed shells), gravel (>0.5–8 cm), rubble (>8–30 cm), small boulders (>30–60 cm), large boulders (≥60 cm), and exposed bedrock (undefined size).

The smallest individuals of *C. frondosa* detected in the video footage measured 1.2 cm in length. Newly-settled recruits and early juveniles (~1 cm or smaller) are typically cryptic, hidden under stones or in crevices and were below the detection threshold of the ROV's camera. Although the detectability of sea cucumbers ≥1.2 cm long was assumed, some may have been obscured from view, such as those partially or completely concealed behind larger boulders or other species.

2.3. Standardization of the virtual quadrat

2.3.1. Parameterization using a reference quadrat

This portion of the study was done to quantitatively determine population densities using a portable (mini-class) ROV where the position of the unit and the camera angle relative to the substratum were not available. In an effort to circumvent this limitation, a reference quadrat (8 × 8 cm) positioned at different angles was photographed with a digital camera (Olympus Tough TG-6) to model the conversion between real-world measurements of this three-dimensional reference quadrat

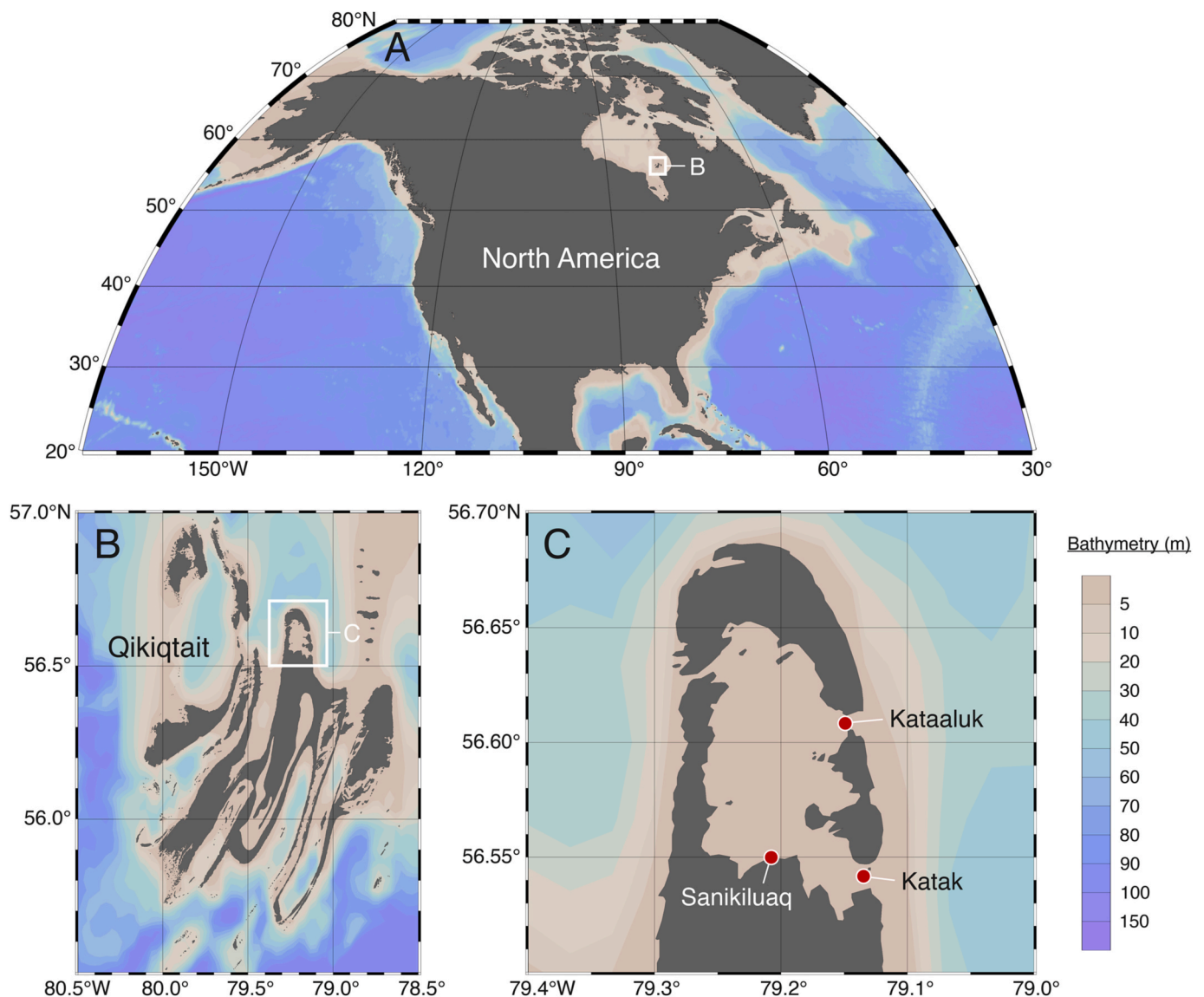


Fig. 1. Sites of ROV deployment for benthic video surveys. (A) Map of North America showing the location of Qikiqtait (boxed). (B) Close up of Qikiqtait in Nunavut, Canada, showing the location of field sites (boxed). (C) Detail of field sites, including Sanikiluaq located off the coast of the hamlet, and channels of Katak and Kataaluk opening into the Hudson Bay. The legend is applicable for panels B and C. Image source: Ocean Data View.

and virtual measurements of two-dimensional images of the same quadrat. In the laboratory, 90 images were taken of this reference quadrat across a range of angles in relation to the digital camera: from directly above, facing downwards at $\sim 90^\circ$ to an angle of $\sim 10^\circ$ (facing forward). Using an image processing software (ImageJ version 1.38), the side of the reference quadrat that was in the foreground of the image (herein referred to as the 'front horizontal length') was fixed to a virtual length of 8 cm corresponding to its real length in the physical world (Fig. 2). This served as the scale to determine the 'vertical midline length' of the reference quadrat in the same image, with a precision of two decimal places (Fig. 2). Next, the 'front interior angle' of one the bottom corners of the reference quadrat in the image was determined (Fig. 2). The proportion between the vertical midline length and the front horizontal length (values ranging from 0 to 1) was computed for each image of the reference quadrat, such that images taken by the digital camera at increasingly perpendicular angle in relation to the reference quadrat were associated with decreasingly lower proportions. This proportion was then plotted against the front interior angle to ascertain the relationship between perspective projection. A non-linear regression was fitted to the plotted data and expressed as an equation.

2.3.2. Application of standardized virtual quadrats

Virtual quadrats (corresponding to 50×50 cm in the physical world) were superimposed onto still frames extracted from ROV videos using a drawing software (Adobe Photoshop version 25.4.0) and an image analysis tool (ImageJ). Still frames were selected at intervals of ~ 5 – 10 s (depending on the speed of the ROV) until the full spectrum of depths was represented. The still frames were only analyzed if they were in focus, the paired lasers were visible, and they showed independent sections of the benthos (i.e., no overlap with next or previous frame selected). For each selected frame, a 'benchmark object' that was located in the background of the image (usually a large non-animate object, such as a rock outcrop) was identified (Fig. 3). To measure the length of the benchmark object, the video footage played forward in time (usually a few frames) until this object was in the horizontal plane of the paired laser points. Returning to the initial image, the known length of the benchmark object in the background was marked. To superimpose a 50×50 cm virtual quadrat, two horizontal lines, representing 50 cm in the physical world, were then drawn: a line intersecting the horizontal plane of the two laser points and a shorter line intersecting the length of the benchmark object (Fig. 3). The known distance between the laser points

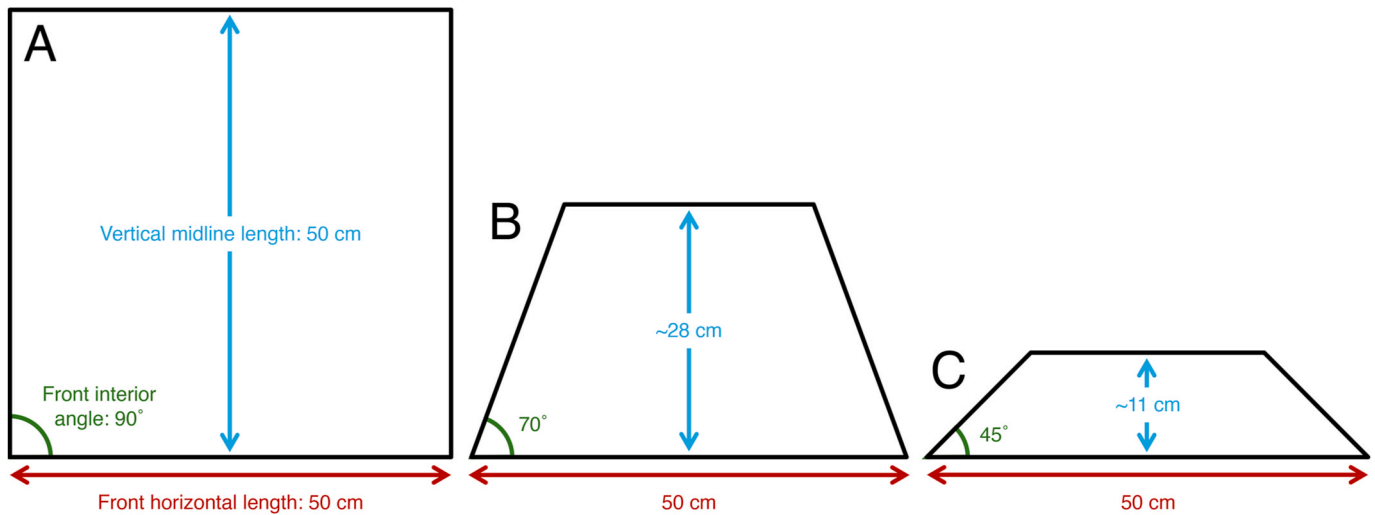


Fig. 2. Schematic of the quadrat viewed with different front interior angles: (A) 90°, (B) 70°, and (C) 45°. The vertical midline lengths were calculated by applying the equation found in Fig. 7 relating the proportion between the vertical midline length and the front horizontal length (i.e., the front edge of the quadrat) with the front interior angle.

(i.e., 12 cm in the physical world) and the known length of the benchmark object were used to draw these two lines. At their respective horizontal planes, these two lines were centered in the image (Fig. 3). Next, two angled vertical lines were drawn to connect the sides of the horizontal lines and extended upward and downward forming the sides of the virtual quadrat (Fig. 3). Should these lines be extended further (beyond the image), they would intersect at the vanishing point. To form the bottom side of the virtual quadrat, a horizontal line was drawn in the foreground (i.e., length of 50 cm in the physical world). To complete the virtual quadrat, (1) the interior angle was determined using ImageJ, (2) the vertical midline length was computed using the non-linear regression equation (see above) and drawn on the image, and (3) a horizontal line was drawn at the top of the midline length, forming the top of the virtual quadrat (Fig. 3).

2.3.3. Validation of quadrat standardization

To validate the application of standardized virtual quadrats and quantify the overall magnitude of distortion due to perspective, we compared the density values between uncorrected and corrected quadrats, superimposed on the same still frame (Fig. 4A and B). Using the paired lasers as a scale, uncorrected quadrats were drawn to dimensions of 50 × 50 cm. Corrected quadrats, representing 50 × 50 cm in the physical world, were standardized using the method described above. For analysis, sets of uncorrected and corrected quadrats of the same still frame with at least one individual of *C. frondosa* were used. The percent change between uncorrected and correct quadrats were computed. A linear regression was applied to assess the relationship between the front interior angle of the corrected quadrat and the percent change in density between uncorrected and corrected quadrats. The front interior angle serves as a proxy for the angle of the camera, i.e., a front interior angle of ~90° represents a camera facing downwards and ~10° represents a camera facing nearly parallel with the seafloor. Uncorrected quadrats associated with a positive percent change in density were categorized as ‘overestimation bias,’ a zero percent change as ‘no bias,’ and a negative percent change as ‘underestimation bias.’ The proportion of these categories of biases was determined for each class of front interior angle (again, a proxy for camera angle), binned in increments of 10°, and for four classes of densities (very low: >0 to <5 ind. m⁻²; low: 5 to <15 ind. m⁻²; intermediate: 15 to <30 ind. m⁻²; and high: 30 to <70 ind. m⁻²).

2.4. Population size structure

To determine population size structure with depth, measurements of unique individuals of *C. frondosa* ($n = 798$ individuals) were made from a series of still frames (image resolution: 2880 × 1625 pixels; $n = \sim 350$ still frames) extracted from video footages over three consecutive days.

The morphometrics of individuals were measured using ImageJ (Supplemental Fig. S1) when two projected laser points were visible near the center of the still frame in the same optical plane as the sea cucumber. Body length (i.e., the length from the base of the crown of tentacles to the posterior end, following the curvature of the body) and, if available, the diameter of the crown of extended tentacles (a straight line from the tip of the tentacles on one side of the crown to the tip of the tentacles on the opposite side) were determined for individuals that were aligned to the horizontal plane of the paired lasers. For the purposes of measurement, tentacles were considered extended when their fine terminal ramifications were visibly oriented and when their main branches (attached to the introvert) were oriented at an oblique angle or nearly perpendicular from the longitudinal axis of the anterior end of the body. The body of some laser-aligned individuals were, however, obstructed in such a way that body length could not be measured. To circumvent this, a linear regression was fitted between tentacle crown diameter and body length ($F_{1,98} = 464$, $p < 0.001$, $R^2 = 0.826$; Supplemental Fig. S2) using individuals for which both measurements were available ($n = 100$ individuals). This relationship ($y = 0.983x + 0.551$) was applied on individuals where only the diameter of the crown of extended tentacles (x) was measured to estimate their body length (y).

2.5. Population density

To determine population density with depth, the number of individuals of *C. frondosa* within a 50 × 50 cm virtual quadrat (0.25 m²) superimposed on a series of still frames ($n = 372$ frames) extracted from video footages (see above) was counted. These still frames were not necessarily the same frames as those used to determine morphometrics of individual sea cucumbers (see above); however, many individuals counted within virtual quadrats were also individuals that were used to determine size structure. Individuals on the edges of the quadrat were only counted if greater than half of their body length was inside the quadrat.

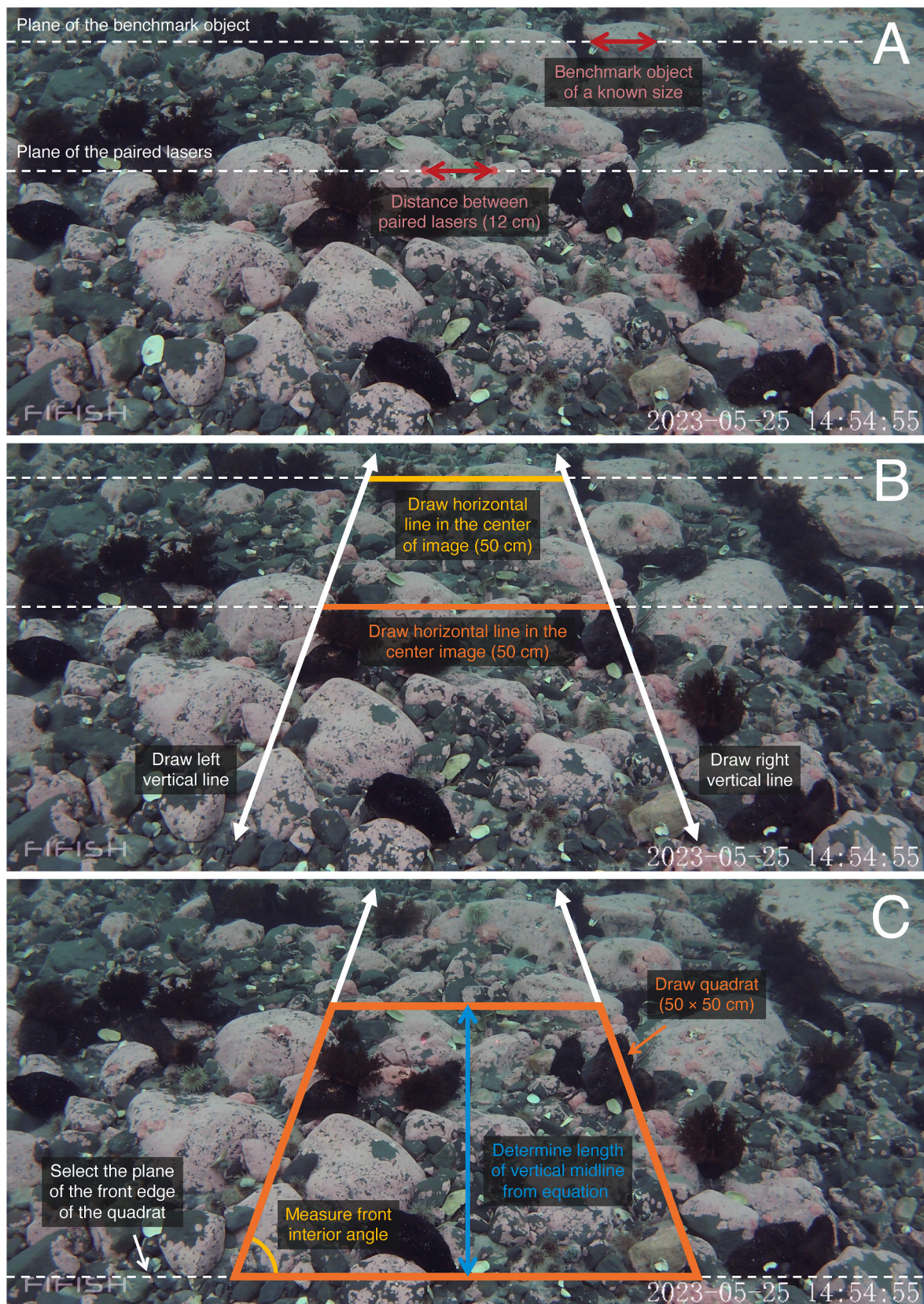


Fig. 3. Steps to superimpose a standardized virtual quadrat (dimensions: 50 × 50 cm) on a still frame. (A) The paired lasers and a benchmark object of a known size in the foreground are indicated in the image. (B) On the planes of the paired lasers and the benchmark object, two horizontal lines, representing 50 cm in the physical world, are drawn at the center of the image; vertical lines on either side are drawn, connecting the aforementioned horizontal lines and extending to the top and bottom of the image. (C) The plane of the front edge of the quadrat is selected and the front interior angle is measured; subsequently, the vertical midline length is computed using the non-linear equation (see Fig. 7); finally, all four sides of quadrat are drawn on the image.

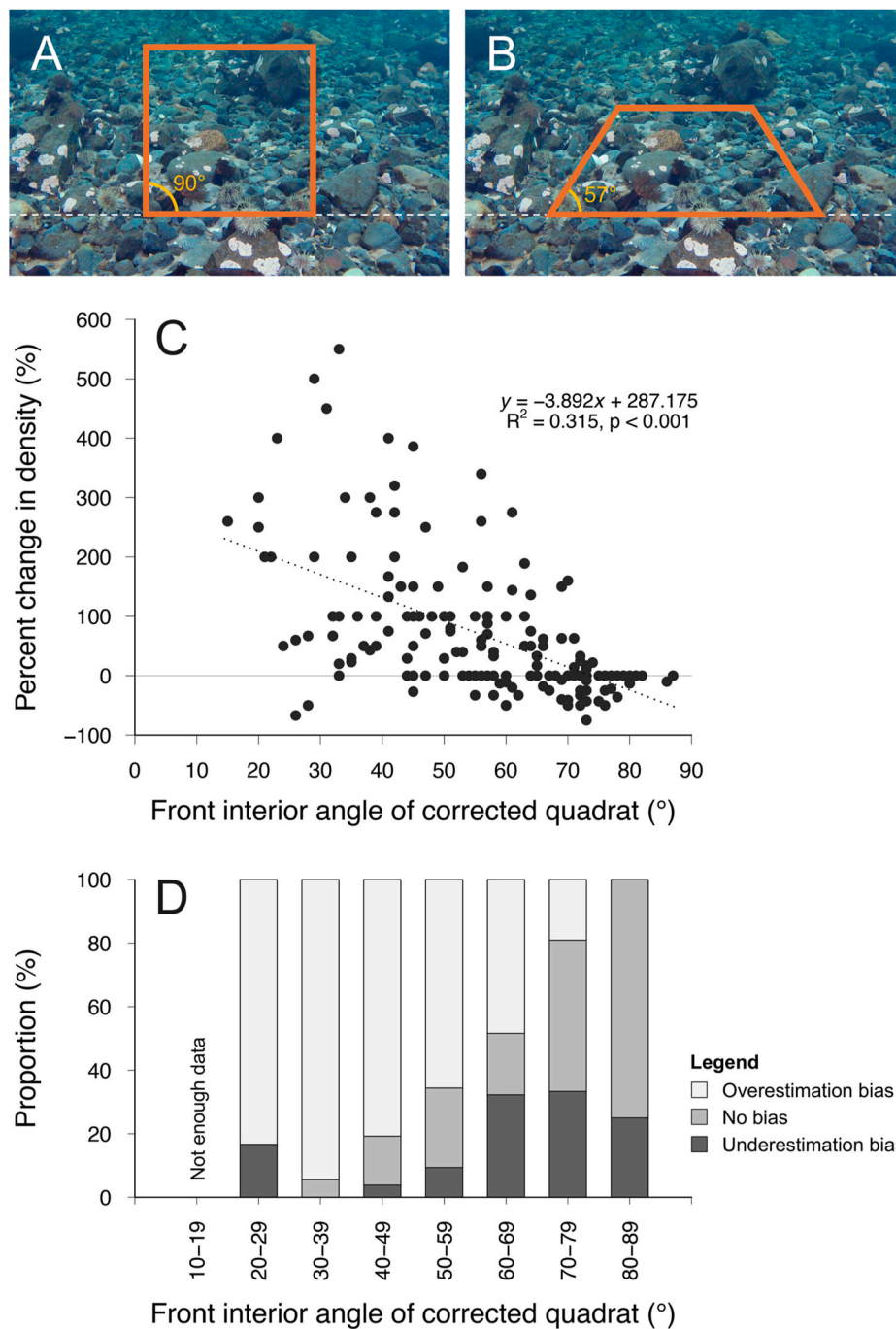


Fig. 4. Distortion in density values due to perspective between corrected and uncorrected quadrats. An example of (A) an uncorrected quadrat (dimensions: 50 × 50 cm using the paired lasers as the only scale for the entire frame) and (B) a corrected quadrat (dimensions: 50 × 50 cm in the physical world), superimposed on the same still frame. (C) Relationship between the front interior angle of the corrected quadrat and the percent change in density between uncorrected and corrected quadrats. (D) Proportion of overestimation bias, no bias, and underestimation for each class of front interior angle (data binned in increments of 10°). Each data point ($n = 170$) represents a pair of correct and uncorrected quadrat with at least one individual of *Cucumaria frondosa*. A front interior angle of 0° represents the orientation which the ROV is parallel to the seafloor (i.e., camera facing forward into the horizon of the seascape) and an angle of 90° represents the orientation which the ROV is perpendicular (i.e., camera facing the seafloor).

2.6. Statistical analyses

Both sets of data on body length and density of *C. frondosa* were not normally distributed (Shapiro–Wilk test, $p < 0.001$). Therefore, a non-parametric Kruskal–Wallis test was used to evaluate variability in body length and, separately, density among sites (pooling all depths together). A post-hoc multiple comparison test (Dunn’s method) was performed when significant effects were detected ($\alpha = 0.017$ after

Bonferroni correction with reported p values adjusted accordingly). To describe bathymetric patterns, the relationships between depth and body length and, separately, between depth and density for each site and for all three sites combined were first evaluated statistically with linear regressions. These relationships were then assessed visually with locally estimated scatterplot smoothing (LOESS) to identify any non-linear trends. Because multiple linear regressions were compared in an analysis, the alpha level was adjusted using the Bonferroni correction to

reduce the risk of type I error (false positives). The LOESS curves were computed with a span of 0.4 except for data from Sanikiluaq for which a span of 0.7 was applied to compensate for the comparatively narrower range of depths. All statistical tests were carried out using the R programming environment (R Core Team, 2024).

3. Results

3.1. Virtual quadrats: analysis of the methodology

Based on data collected from modelling the reference quadrat, the non-linear regression equation $y = 0.000118x^2 - 0.000184x$ was derived, where y is the proportion between the vertical midline length and the front horizontal length and x is the front interior angle of the quadrat ($F_{1,98} = 6510$, $p < 0.001$, $R^2 = 0.985$; Fig. 7). This relationship allowed for the superimposition of a standardized (e.g., a pre-determined area) quadrat on an underwater image of the benthos, provided that there was size-related information for two objects, one towards the foreground (e.g., the laser points) and another towards the background (e.g., the benchmark object).

Comparison between uncorrected (i.e., using the paired lasers as the only scale for the entire frame) and corrected quadrats (i.e., virtual quadrats after standardization) of the same still frame indicated that densities of *Cucumaria frondosa* in uncorrected quadrats tended to overestimate densities. Densities determined using uncorrected quadrats ranged from 4 to 135 ind. m^{-2} (mean $\pm$ standard deviation: 27.3 ± 25.1 ind. m^{-2}) and densities determined using the corresponding set of corrected quadrats ranged from 4 to 68 ind. m^{-2} (mean: 17.8 ± 13.7 ind. m^{-2}). A closer examination of the relationship ($y = -3.892x + 287.175$; $F_{1,168} = 77.2$, $p < 0.001$, $R^2 = 0.315$) showed that perspective-related distortion, measured as percent change in density (y), increased with increasingly forward-facing orientation of the camera, which was quantified by using the front interior angle of the corrected quadrat as a proxy (x) (Fig. 4C).

A breakdown of the data by 10° increments demonstrated that forward-facing camera angles were associated with an 'overestimation bias' and downward-facing angles with 'no bias' (Fig. 4D). More specifically, overestimation bias was the dominant type of bias—i.e., accounting for the largest proportion of uncorrected quadrats—when the corresponding corrected quadrats had a front interior angle of $<70^\circ$ (i.e., a relatively forward-facing camera angle; Fig. 4D). In contrast, no bias dominated when angles were $\geq 70^\circ$ (i.e., relatively downward-facing orientations; Fig. 4D). Lastly, underestimation bias represented the second largest proportion of uncorrected quadrats when the corresponding corrected quadrats had a front interior angle of either $<30^\circ$ or $\geq 60^\circ$ (Fig. 4D).

In addition to camera orientation, the type of bias was influenced by the number of sea cucumbers within the quadrat. Although overestimation bias was detected across all three classes of densities examined, it was the most prevalent (72 %) at intermediate densities (15 to <30 ind. m^{-2} , $n = 54$ quadrats). At very low densities (>0 to <5 ind. m^{-2} ; $n = 38$), 55 % of uncorrected quadrats were unbiased, while 45 % exhibited overestimated bias. At low densities (5 to <15 ind. m^{-2} ; $n = 45$), 51 % of uncorrected quadrats were biased towards overestimating density and 29 % were unbiased. At high densities (30 to <70 ind. m^{-2} ; $n = 33$), 48 % and 42 % of uncorrected quadrats underestimated and overestimated density, respectively.

3.2. Population size structure

Using morphometric measurements collected from each site (depths pooled), there was significant variability in the body lengths of *C. frondosa* among sites ($H = 225.64$, $df = 2$, $p < 0.001$). Mean length ($\pm$ standard deviation) was 6.0 ± 2.7 cm at Sanikiluaq, 7.7 ± 4.5 cm at Katak, and 11.1 ± 2.5 cm at Kataaluk. Sea cucumbers at Kataaluk were larger than those at Sanikiluaq (post hoc test, $Z = 10.03$, $p < 0.001$) and

Katak ($Z = 14.93$, $p < 0.001$) and the latter were larger than those at Sanikiluaq ($Z = 4.52$, $p < 0.001$; Fig. 5A). More precisely, the population size structure at Sanikiluaq was unimodal, consisting of primarily smaller individuals (with the mode centered on 5.2 cm in length). The one at Kataaluk was also unimodal, comprising mainly longer individuals (mode: 11.2 cm). Differently, the one at Katak was bimodal with a mix of both smaller (first mode: 3.5 cm) and larger (second mode: 10.2 cm) size classes (Fig. 5A).

The bathymetric analyses showed no clear relationship between depth and body length at Sanikiluaq (linear regression, $F_{1,337} = 0.6$, $p = 0.430$; Fig. 6A). A negative relationship was detected at Katak ($F_{1,250} = 17.6$, $p < 0.001$, $R^2 = 0.066$), i.e., decreasing size with increasing depth (Fig. 6C). When accounting for the bimodal distribution at this site and omitting the cohort of smaller individuals ingressing to greater depths, a positive relationship was found instead ($F_{1,125} = 19.8$, $p < 0.001$, $R^2 = 0.137$), i.e., increasing size with increasing depth. At Kataaluk, a positive relationship was also detected ($F_{1,205} = 17.2$, $p < 0.001$, $R^2 = 0.077$) (Fig. 6E). Overall, when combining the size data from all three sites (Fig. 6G), a positive relationship between depth and body lengths was present, both with ($F_{1,796} = 17.6$, $p < 0.001$, $R^2 = 0.017$) and without the cohort of small individuals at Katak ($F_{1,667} = 444.7$, $p < 0.001$, $R^2 = 0.400$). All relationships remained statistically significant even after the Bonferroni correction. A closer examination of the global positive relationship using the LOESS curve (pooled data without immature individuals from greater depths) revealed that body lengths tended to increase between ~ 3 and 5 m and then remain relatively constant (~ 12 cm long) downslope (LOESS not shown).

3.3. Population density

Densities of *C. frondosa*, pooled across depths, varied among the three sites surveyed (Fig. 1; $H = 161.32$, $df = 2$, $p < 0.001$). Specifically, density at Sanikiluaq was 24.8 ± 14.7 ind. m^{-2} , which was significantly greater than at Katak (3.4 ± 6.8 ind. m^{-2} ; post hoc test, $Z = -11.85$, $p < 0.001$) and Kataaluk (3.1 ± 5.8 ind. m^{-2} ; $Z = -10.88$, $p < 0.001$; Fig. 5B). No difference in densities was found between Katak and Kataaluk ($Z = -0.29$, $p = 1.000$; Fig. 5B).

An examination of vertical patterns using linear regressions indicated that density increased with increasing depth at Katak ($F_{1,181} = 5.1$, $p = 0.027$, $R^2 = 0.025$) but not at Sanikiluaq ($F_{1,83} = 3.3$, $p = 0.075$) or Kataaluk ($F_{1,102} = 2.2$, $p = 0.145$). At Sanikiluaq, density was low (~ 5 ind. m^{-2}) around 1 m depth, peaked (~ 35 ind. m^{-2}) at ~ 2 m, and tapered to ~ 20 ind. m^{-2} at ~ 3 m (Fig. 6B). At Kataaluk and Katak, sea cucumbers were absent (at least those above the visual detection limit) from the shallowest depths examined down to around 1.5 and 3.5 m depths, respectively. At Katak, descending from shallow waters where no sea cucumbers were detected, two peaks in densities were identified: a smaller peak at ~ 4 m depth, with densities of ~ 7 ind. m^{-2} , and a larger peak at ~ 9 m depth, with densities of ~ 12 ind. m^{-2} (Fig. 6D). At Kataaluk, densities increased comparatively gradually with depth, peaking (i.e., ~ 15 ind. m^{-2}) at depths between 5 and 6 m before dipping (<1 ind. m^{-2}) at ~ 7 m (Fig. 6F). Despite site-specific differences, density decreased with increasing depth when data from all sites were pooled ($F_{1,370} = 28.3$, $p < 0.001$, $R^2 = 0.071$), with a peak in densities at ~ 2 m depth, with densities of ~ 20 ind. m^{-2} (LOESS; Fig. 6H). However, another peak was present at ~ 9 m depth, with densities of ~ 10 ind. m^{-2} (LOESS; Fig. 6H).

3.4. Influence of substratum type

Across all depths examined, sea cucumbers exhibited an affinity for gravel or rubble hard substrata (Figs. 6 and 8). All size-classes of *C. frondosa* were consistently associated with these substratum types (Fig. 8A), which were present at all sites examined. However, the composition of the bottom substrata varied among the sites: Sanikiluaq was predominately composed of rubble (93 %), Katak featured a mix of

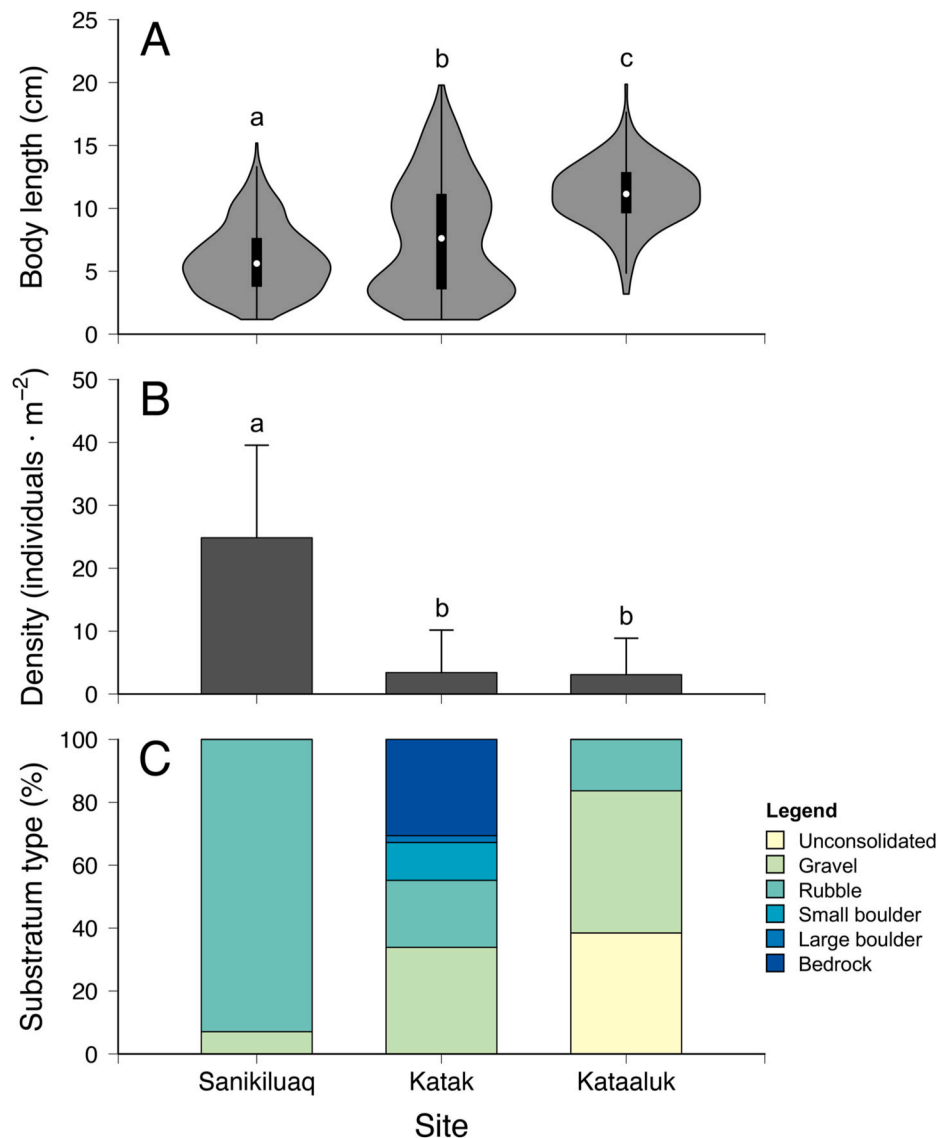


Fig. 5. Overall population size structure and densities of *Cucumaria frondosa* and the distribution of different substrata (all depths pooled). (A) Distribution of body lengths (including estimated body lengths) at Sanikiluaq ($n = 339$ individuals), Katak ($n = 252$), and Kataaluk ($n = 207$). (B) Mean densities at Sanikiluaq ($n = 85$ quadrats), Katak ($n = 183$), and Kataaluk ($n = 104$). (C) Distribution of different substrata at each site by categorizing the substratum type of each quadrat using a modified Wentworth classification scale (Platts et al., 1983) with the addition of ‘unconsolidated substrata’ and ‘exposed bedrock’ categories. Violin plots show the median (white point), interquartile range (thick black bar), and width of the violin shape represents probability (wider = higher probability; thinner = lower probability). Error bars represent (positive) standard deviation. Different letters above the violin plots and bars represent statistically significant differences among sites (post hoc multiple comparisons after Kruskal–Wallis test).

gravel, exposed bedrock, and rubble (34, 31, and 21 %, respectively), and Kataaluk consisted primarily of gravel and unconsolidated substrata (45 and 38 %, respectively) (Fig. 5C). Interestingly, at Katak, exposed bedrock was present at both shallow depths (<4 m) and greater depths (9 and 11 m), yet sea cucumbers were only observed at the deeper range (Fig. 6).

Sea cucumber occupancy patterns reflected the substratum composition of each site. At Sanikiluaq, 94 % of individuals were associated with rubble and the remaining 6 % with gravel. At Katak, 59 % of individuals were found on gravel, 19 % on rubble, 17 % on small boulders, 4 % on bedrock, and 1 % on large boulders. At Kataaluk, 91 % of individuals were associated with gravel and 9 % with unconsolidated substrata.

Globally, substratum preferences were size-dependent. Nearly all (98 %) of the smallest size-class of sea cucumbers (<5 cm in length) utilized gravel or rubble (Fig. 8A). This proportion declined to 69 % for the largest size-class (15 to <20 cm in length; Fig. 8A). While larger

individuals were still predominantly found on gravel and rubble, they were more likely than smaller individuals to occur on bedrock, boulders, and unconsolidated substrata (Fig. 8A). Nevertheless, densities of sea cucumbers on these substrata were typically low (e.g., >0 to <15 ind. m⁻²; Fig. 8B).

The presence and density of *C. frondosa* were also linked to substratum type. Among quadrats with no sea cucumbers, 27 % were associated with gravel, 27 % with rubble, and another 27 % with exposed bedrock (Fig. 8B). Quadrats with low densities of sea cucumbers (>0 to <15 ind. m⁻²) were primarily associated with gravel (47 %; Fig. 8B). In contrast, quadrats with intermediate densities (15 to <30 ind. m⁻²) were predominantly associated with rubble (59 %), as were almost all quadrats (97 %) with high densities (30 to <70 ind. m⁻²) (Fig. 8B).

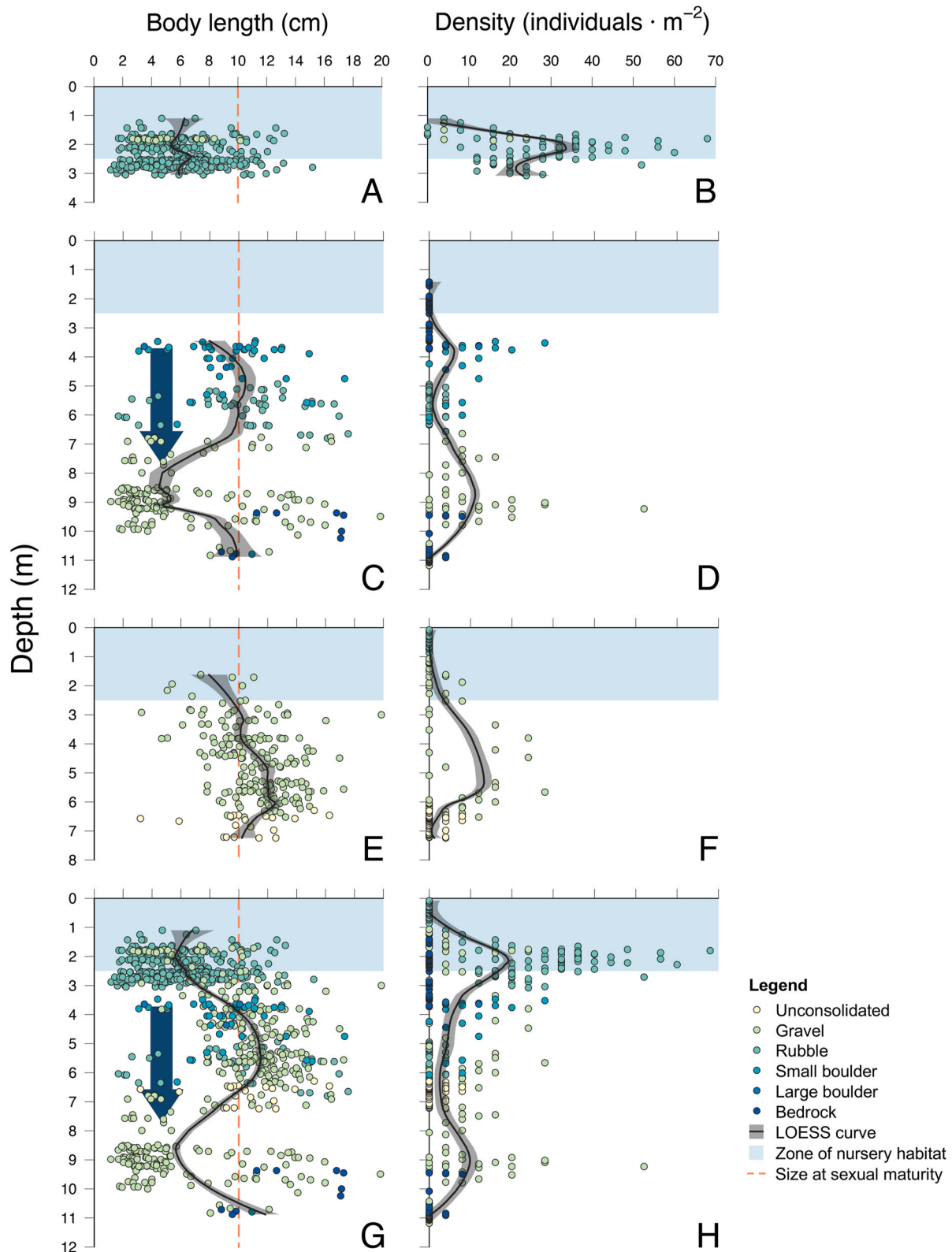


Fig. 6. Vertical distribution of body lengths (including estimated body lengths) and densities of *Cucumaria frondosa* on different substrata. (A–B) Distribution at Sanikiluaq, up to a maximum depth of 3.1 m. (C–D) Distribution at Katak, up to a maximum depth of 11.2 m. (E–F) Distribution at Kataaluk, up to a maximum depth of 7.2 m. (G–H) Data from all three sites pooled together. Arrow (blue) in panels C and G represent the movement of immature individuals from shallow to deeper waters, resulting in an aggregation of individuals at ~9 m depth. Solid black line = locally estimated scatterplot smoothing (LOESS); shaded grey area around the line = standard error; shaded blue area = depth range of the nursery habitat; dashed orange line = approximate size at first sexual maturity (~10 cm in body length). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

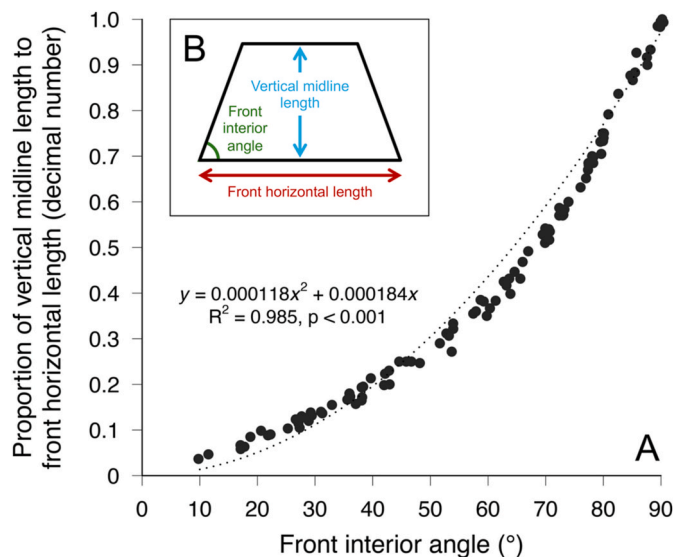


Fig. 7. Non-linear relationship between the front interior angle (x) and the proportion of the vertical midline length to the front horizontal length (y), expressed as a decimal number, of the imaged reference quadrat (n = 100 images) to estimate the vertical midline length when superimposing a virtual quadrat (see Fig. 3). (A) Exponential regression and (B) a schematic of the reference quadrat imaged at an angle.

4. Discussion

Mini-class ROVs offer a cost-effective and versatile alternative to traditional benthic survey methods, including larger work-class ROVs and other non-extractive tools (Montes-Herrera et al., 2021; McGeady et al., 2023, Table 1). As such, they also facilitate capacity building with Indigenous partners (Buscher et al., 2020). Unlike bottom trawls, ROVs have a smaller ecological footprint, and their underwater videos often include paired lasers for scale measurements (Barker et al., 2001; Rochet et al., 2006). However, forward-facing cameras introduce perspective distortion, limiting accurate size measurements of objects along the

horizontal plane of the paired lasers (Wakefield and Genin, 1987). This is why methods using a downward-facing orientation are commonly used in situations where this is possible (e.g., Tyne et al., 2010; Bennett et al., 2016; Bethoney et al., 2019; de Mendonça and Metaxas, 2021).

To take greater advantage of the versatility of mini-class ROVs and address perspective distortions in videos captured with forward-facing cameras at unknown angles, a novel procedure was developed here to superimpose standardized virtual quadrats over still frames. This method enables a more accurate density quantification of benthic taxa without requiring knowledge of the camera angle relative to the sea-floor. Other proposed solutions include relying on four parallel lasers (Tusting and Davis, 1993; Maldonado and Young, 1996; Barker et al., 2001) or a dedicated software that requires extensive calibrations (Bennett et al., 2016). A key advantage of the technique presented here is that it allows analysis using standard two-laser systems, improves maneuverability, and avoids the complexity of tools like the Canadian grid, which requires positional information of the camera and time-intensive calculations (Wakefield and Genin, 1987).

Using this novel approach, the present study provided a first characterization of the size and density distributions of the sea cucumber *Cucumaria frondosa* at depths covering nursery habitats down to ~2.5 m depth described in Hamel et al. (2023), and further downslope to ~11.5 m, in a region of the Canadian Arctic where dive surveys are, at present, logistically challenging. The composition of the substratum types and the spatial distribution of body lengths (individuals ≥ 1.2 cm, minimum detectable size on videos) and densities of *C. frondosa* revealed differences among sites and depths, highlighting geographic variability in addition to bathymetric particularities.

It had been established previously in the same geographic areas that newly-settled recruits (~0.5–10 mm long) to early juveniles (up to ~10 mm long) occurred among mussels and under stones (Hamel et al., 2023; Ma et al., 2025). These nursery grounds stretched from the shore down to 2.5 m in areas that can be seasonally subjected to salinities as low as 11 psu, a condition with which older juveniles and mature individuals cannot cope (J.-F. Hamel, unpublished data). Living in intermittent shallow hyposaline habitats close to the interface with sea ice might afford new recruits and early juveniles protection against less tolerant predators that were observed in higher densities in deeper strata. Experimental studies showed that once *C. frondosa* reaches a length of

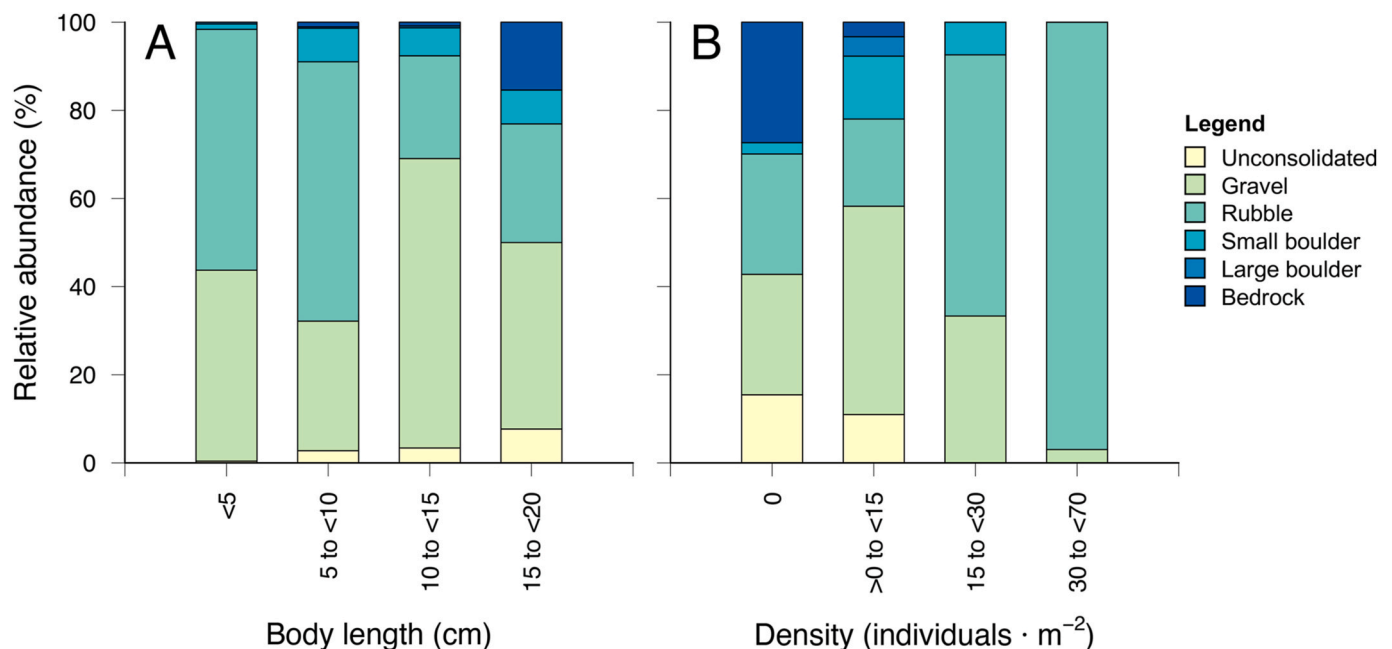


Fig. 8. Distribution of different substrata associated with different classes of (A) body lengths (including estimated body lengths) and (B) densities of *Cucumaria frondosa* (all sites and depths pooled).

Table 1

Summary of image-based, non-destructive, non-extractive methods for proximal characterization of the benthic environment, excluding more destructive image-based methods, such as bottom trawling and towed camera systems. Remote sensing methods, such as crewed aerial surveys and unoccupied aerial vehicles (or drones) were also excluded.

Method	Cost (USD)	Training	Advantages	Limitations
Diver-operated camera	\$100s to \$1,000s.	Moderate to high: Up-to-date training required as a SCUBA diver. Knowledge of underwater photography.	Diver can place a physical quadrat within the frame of the image, without the need to manually calculate area. Multiple images of the same quadrat can be taken. Allows for <i>in situ</i> adjustments in real-time.	Limited to practical depths of 30–50 m that divers can safely reach. Facilities to service dive equipment and ensure safety must be accessible, making remote environments particularly challenging. Time and resource-intensive, requiring multiple dives/divers for good spatial coverage.
Drop camera system	\$1,000s to \$10,000s.	Low to moderate: Some expertise required to deploy and retrieve the system. Accessorily, this method requires well-trained vessel operators.	Tilt of the camera and distance from the substratum are fixed, allowing the area surveyed to be easily determined.	Requires a launching platform (vessel) in open ice-free waters. Limited maneuverability: Although greater depths can be reached compared to divers, the system is difficult to operate under strong currents and only small spatial coverage is possible near ice edges or under ice.
Mini-class remotely-operated vehicle (ROV)	\$1,000 to \$10,000s.	Low: Can be operated by almost anyone with minimal training.	Mini-class ROVs are cost-efficient, highly manoeuvrable, and easily launched under sea ice through an opening. Can survey at greater depths (typically up to 100s of m) than SCUBA.	Positional information not available and can be tilted by currents; spatial coverage limited by battery life and length of tether.
Work-class remotely-operated vehicle (ROV)	\$100,000s to ≥\$1,000,000s, depending on features.	High to very high: Formal training required for piloting; accessorily, this method requires well trained vessel operators.	Can survey at greater depths (typically up to 1,000s of m) than mini-class ROVs, carry heavy payloads, and be equipped with specialized tools (e.g., cutting tools).	Costly to safely operate; requiring a platform (vessel or wharf) and specialized equipment (Launch and Recovery System; LARS) to launch and retrieve.
Autonomous underwater vehicle (AUV)	≥\$100,000, depending on features.	Very high: Specialized training required in programming and mission planning. Technical expertise required in maintenance and repairs.	Untethered, can survey deep and remote regions. Autonomous operation: Can stay submerged for long periods (e.g., up to six months) and survey large areas without human intervention. Can resurvey the same area repeatedly with a programmed route.	Although costly to purchase, operate, and maintain, smaller low-cost AUVs may become more widely available. Deployments limited by battery life. Susceptible to navigational and positional errors, increasing the risk of equipment loss or failure.
Unmanned surface vehicle (USV)	\$50,000 to \$100,000s.	High: Navigational knowledge and training in programming required.	Autonomous operation. Data can be integrated with geographic information (GPS coordinates).	Operationally limited to clear and shallow waters that are not obstructed by sea ice. Difficult to operate in rough weather or high wave conditions.

~10 mm, its capacity to move around increases, presumably allowing it to transition from a sheltered existence to one where it can be exposed at the surface of the substratum (Hamel and Mercier, 1996a). The present study confirmed for the first time in the wild this safe size at which juvenile can emerge from cryptic habitats. Broadly speaking, the distributional patterns reported here in Qikiqtait support that this emergence is followed by a gradual downslope ontogenetic migration out of the nursery habitats. The findings also align with previous observations on *C. frondosa* from other regions of its geographic distribution. In eastern Canada, smaller individuals (~10–20 mm long) were found in shallower water (0–20 m) and larger individuals dominated at the maximum depth surveyed of 40–60 m (Hamel and Mercier, 1996a). Morrison et al. (2025) showed that the population size structure in deeper water around Qikiqtait (>30 m) almost exclusively harbored individuals >10 cm long, reinforcing the probable ontogenetic migration with depth in the species.

The preference for gravel, stones, and other hard substrata in the present study also found echoes in other studies on *C. frondosa* conducted both in Newfoundland and eastern Québec (Hamel and Mercier, 1996a; So et al., 2010). Fairly high hydrodynamic conditions characterize the sites in which *C. frondosa* can be found (see review by Gianasi et al., 2021). In fact, it was demonstrated under laboratory conditions that *C. frondosa* avoids calm areas and actively migrates towards zones with intermediate current speeds of ~40 cm s⁻¹ (Sun et al., 2018). It was also seen to most actively feed where currents were between 55 and 130 cm s⁻¹ (Holtz and MacDonald, 2009). This could explain higher densities recorded here at certain sites or depths (e.g., the shallow shoal) where current conditions may be optimal. Moreover, individuals from all surveyed areas in the present study were typically found perched on the stones, supporting their need to be exposed to turbulent water, as also seen by Morrison et al. (2025).

Assessment of size distributions for body lengths revealed that individuals at Kataaluk were larger than those at Katak and Sanikiluaq, with the latter site generally harboring the smallest individuals. While it did not differ between the two channel sites (Katak and Kataaluk), the density of sea cucumbers was higher at the shallow shoal site (Sanikiluaq), where the aggregation of smaller individuals (~5 cm long) may reflect pulses of settlement on the nearby dominant rubble substratum. At depths down to ~6 m, there were similarities with respect to the size distributions at the two channel sites, indicating a transition of smaller individuals (~5 cm long) to larger ones (~10 cm, assumed to be the size at first sexual maturity; Hamel and Mercier, 1996a). This further supports a pattern of downslope migration, with individuals transitioning through sexual maturity between ~3 m and ~5 m depths.

In slight contrast, the bimodal population at Katak, with a conspicuous aggregation of predominantly immature individuals (<5 cm) at intermediate depths (~9 m), suggests a unique habitat. This could be a result of topographic or substratum features, since Katak had a more complex rubble/boulder/bedrock composition relative to the other sites, modulating habitat suitability, food availability, and predation pressure. Some of the predators of *C. frondosa* (Legault and Himmelman, 1993; Hamel and Mercier, 1996a; Francour, 1997; So et al., 2010), particularly the sea star *Solaster endeca* preying on both adults and juveniles and the sea urchin *Strongylocentrotus droebachiensis* grazing solely on recruits and small juveniles, were present in the general study area. The concentration of smaller individuals of *C. frondosa* at ~9 m depth in Katak may be linked to a local absence of these predators, and/or the prevalence of shelters offered by the complex substrata. Incidentally, the predatory sea star *S. endeca* was only observed at greater depths (K.C.K. Ma, personal observation). Another contributor may be the proximity of Katak to the shoal of Sanikiluaq, which hosts a large source population of juveniles (this study). Overall, the bimodal

size distribution at Katak suggests that immature individuals may congregate until they reach maturity, possibly resulting in pulses of downslope migration at specific stages of development as hypothesized for populations in the St. Lawrence Estuary (Québec) thirty years ago by Hamel and Mercier (1996a). Migration into greater depths can rely on active crawling, although this mode of dispersal limits adult sea cucumber to speeds of $\sim 1.2 \text{ m d}^{-1}$ (Hamel and Mercier, 1996a; Gianasi et al., 2018). However, migration in *C. frondosa* (newly-settled recruits, juveniles, and adults) can be achieved through active buoyancy adjustment (ABA) whereby individuals become neutrally/positively buoyant and can either drift with the prevailing current or roll down a slope (Hamel et al., 2019). Arguably, larger mature individuals eventually move into deeper waters beyond the maximum depth investigated in this study (i.e., the species is known to occur down to at least 300 m; Mercier et al., 2023), where conditions are likely to be more stable. The presence of a slope (e.g., at the channel sites) may favor vertical movement of sea cucumbers into deeper waters more readily than flatter bottoms (e.g., at the shoal site) where retention may be occurring, in line with the higher population density measured at Sanikiluaq. Salinity drops, such as may be expected in shallower areas, are known to trigger ABA under laboratory conditions (Hamel et al., 2019), representing a potential trigger of seasonal (spring) downslope migrations. Future studies should explore ecological processes driving the geographic and bathymetric distribution in the population size structure and density of *C. frondosa*.

Densities of *C. frondosa*, across different depths examined, varied between ~ 3 and 35 ind. m^{-2} in the three studied sites. In the Canadian Arctic, a recent review of the literature highlighted that the density of this sea cucumber was variable across the Nunavut territory (Hamel and Mercier, 2024). More precisely, densities around 0.5 ind. m^{-2} were recorded for deeper nearby sites in Qikiqtait (Morisson et al., 2025), in line with the various values reported in the literature for other locations in its distribution range, including the Estuary and Gulf of St. Lawrence (i.e., 0.05 to $>5 \text{ ind. m}^{-2}$; Hamel and Mercier, 1996a; Campagna et al., 2005; Dallaire et al., 2013). Dredge surveys also revealed densities between 0.6 and 0.7 ind. m^{-2} in Icelandic fishing zones (MFRI, 2019). However, channels and other nearby sites in Qikiqtait can host adult densities as high as 50 ind. m^{-2} (Hamel and Mercier, 2024).

Basic biological and ecological knowledge is vital for the protection of species of commercial interest. Monitoring size and density distributions of *C. frondosa* in coastal waters of the Canadian Arctic is critical for developing sustainable management practices in support of subsistence harvests and of proposed small-scale commercial expansions. This could be done using mini-class ROVs to help focus efforts on areas where larger individuals occur at higher densities while safeguarding nursery habitats and populations of spawning individuals through conservation initiatives. Globally, true nursery habitats for sea cucumbers are seldom identified (reviewed by Wolfe et al., 2024). The new recruits of *Holothuria scabra* were found on seagrass leaves in Solomon Islands (Mercier et al., 2000), implying an ontogenetic migration to nearby muddy substrata, while those of *H. floridana* occurred near or on adults in Belize (Rogers et al., 2021). More broadly, juvenile echinoderms are poorly studied, but their unique ecology can be instrumental to the maintenance of populations, as exemplified by the resilient herbivorous ‘juveniles-in-waiting’ of the crown-of-thorns sea star *Acanthaster planci*, which play a key part in outbreaks of the corallivorous adults (Foo et al., 2024). For commercially-important species, identifying and protecting nursery grounds is crucial due to their important role in maintaining healthy stocks. The present findings bring additional support to the downslope migration of maturing individuals as a contributor to the complex spatial dynamics of *C. frondosa*, underscoring the need for spatially-explicit management strategies.

CRedit authorship contribution statement

Kevin C.K. Ma: Writing – review & editing, Writing – original draft,

Visualization, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Lauren E. Gover:** Writing – review & editing, Writing – original draft, Visualization, Investigation, Formal analysis, Data curation. **Jean-François Hamel:** Writing – review & editing, Project administration, Investigation, Conceptualization. **Annie Mercier:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.polar.2025.101287>.

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